Essential in Hypermesh (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	Introduction to FEM Brief on Meshing Basic interaction with Hypermesh User interface Opening/saving files Working with panels Model organization Display control Preparing geometry for meshing Creating and Editing Line Data, Creating and Editing Solid Geometry Importing and Repairing CAD	Theory/Demo	2.0Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0Hrs
3	Session 3	Mid surfaces Simplifying Geometry Refining Topology Geometry and Mesh Editing using Quick Edit Panel	Theory/Demo	2.0Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0Hrs
5	Session 5	Shell meshing **Auto meshing:** meshing on surface geometry Meshing without surfaces 2D Mesh in curved QI Mesh Creation	Theory/Demo	2.0Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0Hrs
7	Session 7	Meshing a Model using Shrink Wrap Tetra meshing **Method 1 –** Standard Tetramesh **Method 2 –** Volume Tetramesh	Theory/Demo	2.0Hrs
8	Session 8	Exercises included for Practice	Practicals	2.0Hrs
9	Session 9	Checking tetra element quality Remeshing tetra elements Creating hexa and penta mesh Creating hex-penta mesh using surfaces	Theory/Demo	2.0Hrs
10	Session 10	Exercises included for Practice	Practicals	2.0Hrs
11	Session 11	Creating hexahedral mesh with the solid map function Quality Checking and Editing Mesh	Theory/Demo	2.0Hrs
12	Session 12	Exercises included for Practice	Practicals	2.0Hrs
13	Session 13	**Assemblies:** welding and swapping parts Spot connectors Area connectors Bolt connectors Part replacement	Theory/Demo	2.0Hrs
14	Session 14	Exercises included for Practice	Practicals	2.0Hrs
15	Session 15	Analysis Setup Setting up loading conditions Formatting Model for analysis Obtaining and assigning beam cross section properties using Hyper Beam	Theory/Demo	2.0Hrs
16	Session 16	Exercises included for Practice	Practicals	2.0Hrs
17	Session 17	Defining composites Working with loads on geometry	Theory/Demo	2.0Hrs
18	Session 18	Exercises included for Practice	Practicals	2.0Hrs
19	Session 19	Hypermesh Solver Interfaces Review, Test and Project Discussion	Theory/Demo	2.0Hrs
20	Session 20	Project Workbook	Practicals	2.0Hrs